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Project      : Sales Analysis using SQL
File Name    : 05_Business_Analysis.sql
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Description :
This script answers key business questions using SQL.
It uses joins, aggregate functions, Common Table Expressions (CTEs),
and Window Functions to generate business insights.

Business Questions:
1. Product Category Performance
2. Regional Performance
3. Top 5 Sales Representatives
4. Monthly Sales Trend Analysis
5. Revenue Contribution by Region
6. Top Product Category in Each Region
7. Month-over-Month Revenue Growth

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=====
-- Business Question 1: Product Category Performance
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=====
SELECT
    p.product_category,
    ROUND(SUM(f.sales_amount) / 1000000, 2) AS total_revenue_million,
    SUM(f.quantity_sold) AS total_quantity_sold,
    ROUND(AVG(f.sales_amount), 2) AS average_sales_amount
FROM fact_sales f
JOIN dim_product p
    ON p.product_key = f.product_key
GROUP BY p.product_category
ORDER BY total_revenue_million DESC;

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=====
-- Business Question 2: Regional Performance
--
=====
SELECT
    r.region,
    ROUND(SUM(f.sales_amount) / 1000000, 2) AS total_revenue_million,
    COUNT(f.sales_key) AS total_transactions,
    ROUND(SUM((f.unit_price - f.unit_cost) * f.quantity_sold), 2) AS

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total_profit,
    ROUND(
        SUM((f.unit_price - f.unit_cost) * f.quantity_sold)
        / SUM(f.sales_amount) * 100,
        2
    ) AS profit_margin_percentage
FROM fact_sales f
JOIN dim_region r
    ON r.region_key = f.region_key
GROUP BY r.region
ORDER BY total_revenue_million DESC;

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-- Business Question 3: Top 5 Sales Representatives
--
=====
SELECT
    sr.sales_rep,
    ROUND(SUM(f.sales_amount) / 1000000, 2) AS total_revenue_million,
    ROUND(SUM((f.unit_price - f.unit_cost) * f.quantity_sold), 2) AS
total_profit,
    ROUND(
        SUM((f.unit_price - f.unit_cost) * f.quantity_sold)
        / SUM(f.sales_amount) * 100,
        2
    ) AS profit_margin_percentage,
    COUNT(f.sales_key) AS total_transactions,
    SUM(f.quantity_sold) AS total_quantity_sold
FROM fact_sales f
JOIN dim_sales_rep sr
    ON sr.sales_rep_key = f.sales_rep_key
GROUP BY sr.sales_rep
ORDER BY total_revenue_million DESC
LIMIT 5;

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-- Business Question 4: Monthly Sales Trend Analysis
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SELECT
    d.year,
    d.month,
    d.month_name,
    ROUND(SUM(f.sales_amount) / 1000000, 2) AS total_revenue_million,
    ROUND(SUM((f.unit_price - f.unit_cost) * f.quantity_sold), 2) AS
total_profit,
    ROUND(
        SUM((f.unit_price - f.unit_cost) * f.quantity_sold)
        / SUM(f.sales_amount) * 100,

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        2
    ) AS profit_margin_percentage,
    COUNT(f.sales_key) AS total_transactions,
    SUM(f.quantity_sold) AS total_quantity_sold
FROM fact_sales f
JOIN dim_date d
    ON d.date_key = f.date_key
GROUP BY d.year, d.month, d.month_name
ORDER BY d.year, d.month;

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=====
-- Business Question 5: Revenue Contribution by Region
--
=====
WITH company_revenue AS (
    SELECT SUM(sales_amount) AS total_revenue
    FROM fact_sales
)

SELECT
    r.region,
    ROUND(SUM(f.sales_amount) / 1000000, 2) AS total_revenue_million,
    ROUND(
        SUM(f.sales_amount) / cr.total_revenue * 100,
        2
    ) AS revenue_contribution_percentage
FROM fact_sales f
JOIN dim_region r
    ON r.region_key = f.region_key
CROSS JOIN company_revenue cr
GROUP BY r.region, cr.total_revenue
ORDER BY revenue_contribution_percentage DESC;

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-- Business Question 6: Top Product Category in Each Region
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WITH ranked_products AS (

    SELECT
        r.region,
        p.product_category,
        ROUND(SUM(f.sales_amount) / 1000000, 2) AS total_revenue_million,

        DENSE_RANK() OVER(
            PARTITION BY r.region
            ORDER BY SUM(f.sales_amount) DESC
        ) AS revenue_rank

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FROM fact_sales f
JOIN dim_region r
    ON r.region_key = f.region_key
JOIN dim_product p
    ON p.product_key = f.product_key

GROUP BY
    r.region,
    p.product_category

)

SELECT *
FROM ranked_products
WHERE revenue_rank = 1;

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=====
-- Business Question 7: Month-over-Month Revenue Growth
--
=====
WITH monthly_revenue AS (

    SELECT

        d.year,
        d.month,

        ROUND(
            SUM(f.sales_amount) / 1000000,
            2
        ) AS total_revenue_million,

        LAG(
            ROUND(SUM(f.sales_amount) / 1000000, 2)
        ) OVER(
            PARTITION BY d.year
            ORDER BY d.month
        ) AS previous_month_revenue

    FROM fact_sales f
    JOIN dim_date d
        ON d.date_key = f.date_key

    GROUP BY
        d.year,
        d.month

)

SELECT

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year,  
month,  
total_revenue_million,  
previous_month_revenue,  
  
ROUND(  
    total_revenue_million - previous_month_revenue,  
    2  
) AS revenue_growth,  
  
ROUND(  
    (  
        (total_revenue_million - previous_month_revenue)  
        / NULLIF(previous_month_revenue, 0)  
    ) * 100,  
    2  
) AS revenue_growth_percentage  
  
FROM monthly_revenue;
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